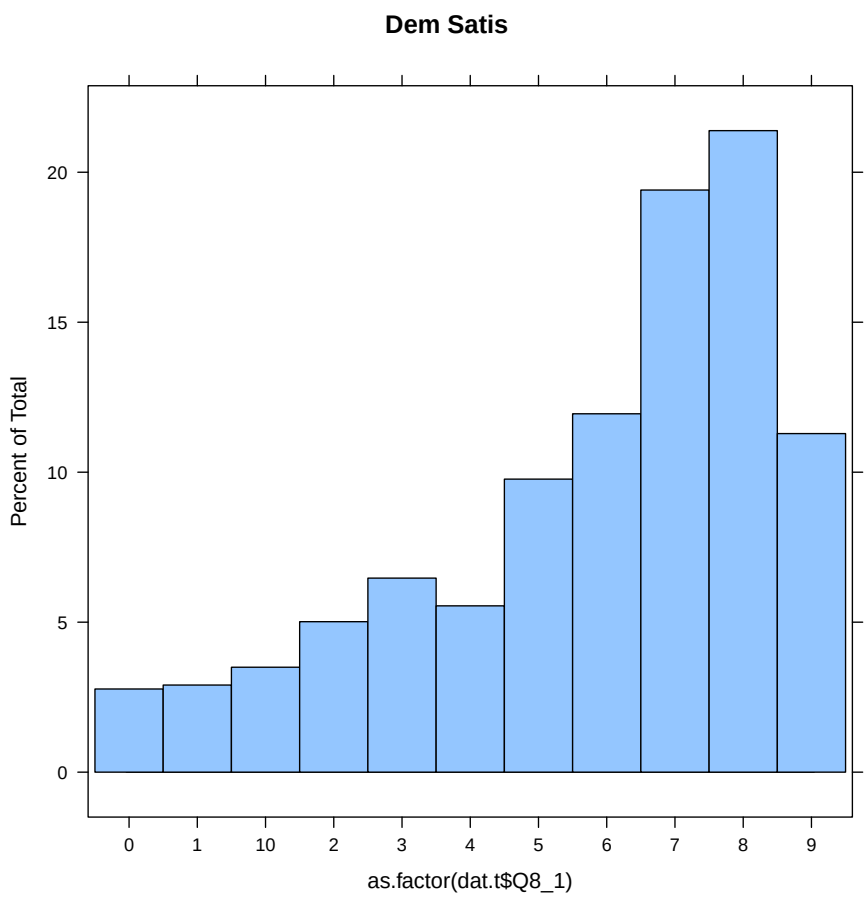
```r
# load data
load(file="/Users/hectorbahamonde/research/democratic_backsliding/2025/dat_t.RData")
# Pacman
if (!require("pacman")) install.packages("pacman"); library(pacman)

## Loading required package:  pacman
```

Sample size is n = 1515

```r
table(dat.t$Q5) # Geographical distribution

##
##         Helsinki and Uusimaa Northern and Eastern Finland
##                          449                          361
##             Southern Finland              Western Finland
##                          322                          383

table(dat.t$Q6) # Education

##
##        Less than primary school (grades 1-9)
##                                            5
##                    Matriculation examination
##                                          203
##                             Other/Don't know
##                                           12
##                  Primary school (grades 1-9)
##                                          109
##                   Professional qualification
##                                          545
##                            University degree
##                                          282
## Upper secondary school or polytechnic degree
##                                          359

table(dat.t$Q3) # Age

##
## 18-24 25-34 35-44 45-54   55+
##   169   272   261   276   537

table(dat.t$Q4) # gender

##
##                   Female                     Male Other/do not want to say
##                      763                      747                        5

lattice::histogram(as.factor(dat.t$Q8_1), main = "Dem Satis") # Dem satisf
```




```
table(dat.t$Q10_1) # Democracy may have its problems, but it is better than other forms of g

##
##                I don't know Neither agree nor disagree
##                          43                        190
##              Somewhat agree          Somewhat disagree
##                         494                         71
##               Totally agree           Totally disagree
##                         682                         35

table(dat.t$Q10_2) # Democracy is not an effective form of government, and it would be bette

##
##                I don't know Neither agree nor disagree
##                          37                        190
##              Somewhat agree          Somewhat disagree
```

```
##                        164                        334
##              Totally agree           Totally disagree
##                        107                        683
```

```
table(dat.t$Q10_3) # It is important that Finland has a free and politically independent me
```

```
##
##               I don't know Neither agree nor disagree
##                         23                        127
##             Somewhat agree          Somewhat disagree
##                        273                         36
##              Totally agree           Totally disagree
##                       1028                         28
```

```
table(dat.t$Q10_4) # The right of citizens to political participation should not be restric
```

```
##
##               I don't know Neither agree nor disagree
##                         38                        154
##             Somewhat agree          Somewhat disagree
##                        481                         70
##              Totally agree           Totally disagree
##                        734                         38
```

```
table(dat.t$Q10_5) # There should be no political pressure on the judiciary
```

```
##
##               I don't know Neither agree nor disagree
##                         67                        213
##             Somewhat agree          Somewhat disagree
##                        353                         88
##              Totally agree           Totally disagree
##                        744                         50
```

```
table(dat.t$Q11) # Interest in politics
```

```
##
##             I don't know    Not at all interested Only a little interested
##                        8                      102                      308
##            Some interest          Very interested
##                      689                      408
```

```
table(dat.t$Q15) # Voted for
```

```
##
##           Centre Party of Finland (KESK)
##                                       89
```

```
##      Christian Democrats of Finland (KD)
##                                       32
##                    Green Alliance (VIHR)
##                                      119
##                           I did not vote
##                                      235
##                             I don't know
##                                       58
##                      Left Alliance (VAS)
##                                      107
##                             Movement Now
##                                       16
##           National Coalition Party (NCP)
##                                      173
##                  Other party or grouping
##                                       22
## Social Democratic Party of Finland (SDP)
##                                      317
##  Swedish People's Party of Finland (RKP)
##                                       31
##                           True Finns (PS)
##                                      316
```

```
table(dat.t$Q13_1) # The government must be able to censor media that are unduly critical o
```

```
##
##               I don't know Neither agree nor disagree
##                         59                        181
##             Somewhat agree          Somewhat disagree
##                        161                        229
##              Totally agree           Totally disagree
##                         74                        811
```

```
table(dat.t$Q13_2) # The right to demonstrate must be protected, even if it makes life diff
```

```
##
##               I don't know Neither agree nor disagree
##                         54                        263
##             Somewhat agree          Somewhat disagree
##                        459                        243
##              Totally agree           Totally disagree
##                        366                        130
```

```
table(dat.t$Q13_3) # The government should be able to ignore court decisions that it conside
```

```
##
##               I don't know Neither agree nor disagree
```

```
##                        197                        374
##             Somewhat agree          Somewhat disagree
##                        289                        229
##              Totally agree           Totally disagree
##                        150                        276
```

```
table(dat.t$Q13_4) # Members of parliament should be able to question and scrutinise governm
```

```
##
##               I don't know Neither agree nor disagree
##                         71                        257
##             Somewhat agree          Somewhat disagree
##                        565                        131
##              Totally agree           Totally disagree
##                        426                         65
```

```
table(dat.t$Q13_5) # The government should be able to stretch the limits of legislation to s
```

```
##
##               I don't know Neither agree nor disagree
##                         97                        292
##             Somewhat agree          Somewhat disagree
##                        432                        253
##              Totally agree           Totally disagree
##                        252                        189
```

```
table(dat.t$Q13_6) # Our country would work better if political decisions were left to indep
```

```
##
##               I don't know Neither agree nor disagree
##                        118                        383
##             Somewhat agree          Somewhat disagree
##                        436                        248
##              Totally agree           Totally disagree
##                        197                        133
```

```
table(dat.t$Q13_7) # Our country would work better if political decisions were left to succe
```

```
##
##               I don't know Neither agree nor disagree
##                        101                        296
##             Somewhat agree          Somewhat disagree
##                        265                        328
##              Totally agree           Totally disagree
##                         95                        430
```

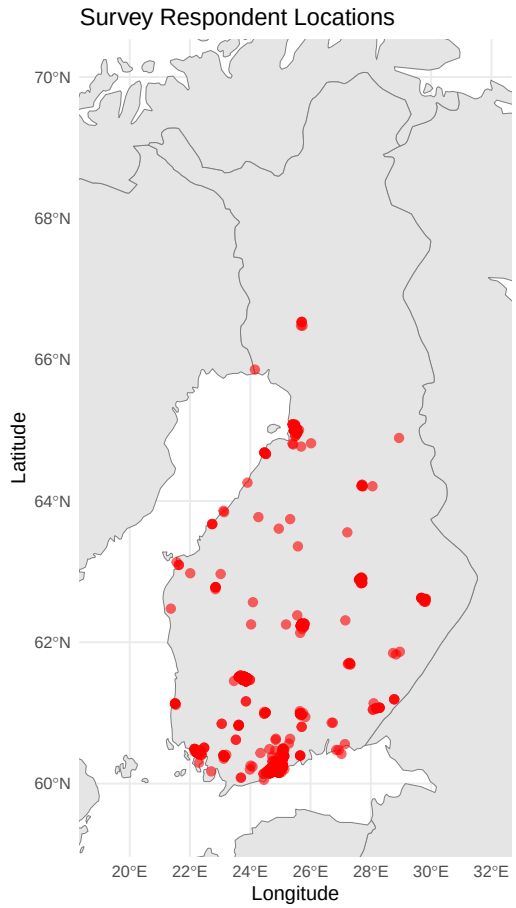
```r
# Don't knows ISSUE
library("tidyverse")

## -- Attaching core tidyverse packages ------------------------ tidyverse
2.0.0 --
## v dplyr     1.1.4     v readr     2.1.5
## v forcats   1.0.0     v stringr   1.5.1
## v ggplot2   3.5.1     v tibble    3.2.1
## v lubridate 1.9.4     v tidyr     1.3.1
## v purrr     1.0.2
## -- Conflicts ------------------------------------------ tidyverse_conflicts()
--
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to
force all conflicts to become errors

dont_know_variations <- c("Other/Don't know", "I don't know")

# Count the occurrences of "don't know" responses per respondent
library("tidyverse")
dont_know_count <- dat.t %>%
  select(-ResponseId) %>%  # Exclude the ResponseId column
  mutate_all(~ . %in% dont_know_variations) %>%  # Check for "don't know" variations
  rowSums()  # Count "don't know" responses per respondent

# Combine with ResponseId for reference
respondent_dont_know <- data.frame(
  ResponseId = dat.t$ResponseId,
  DontKnowCount = dont_know_count
)

# Identify the respondents with the most "don't know" responses
most_dont_know <- respondent_dont_know %>%
  arrange(desc(DontKnowCount)) %>%
  filter(DontKnowCount > 0)  # Optional: Exclude respondents with no "don't know" answers
```

Currently, there are 136 respondents (identified below) that rank pretty high with their don't know" answers. For example subject $R_8FeJf5YKxHQEolE$ ranks 53 don't knows.

```r
head(most_dont_know)

##          ResponseId DontKnowCount
## 1 R_8FeJf5YKxHQEolE            53
## 2 R_2R3knXLJ4NC90Jw            49
```

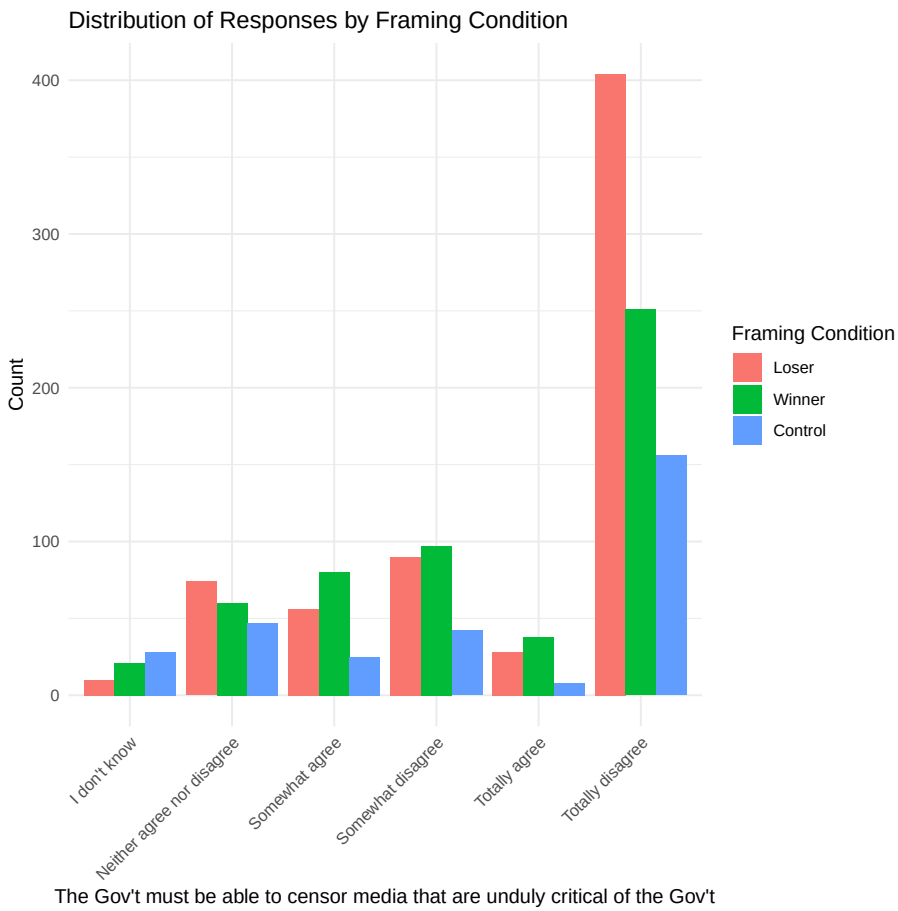
```
## 3 R_8gLtg6j9gLHuw35            45
## 4 R_2kHzTpObvBZCZL9            44
## 5 R_8xxDb4wqdBqi5D6            42
## 6 R_8D8E0aB5j69L0Zw            42
```

```
### Map of respondents
# setting "<NA>" to NA
p_load(dplyr)
dat.t <- dat.t %>% mutate(across(where(is.character), ~ ifelse(. == "<NA>", NA, .)))


dat.t <- dat.t %>%
  mutate(
    LocationLatitude = as.numeric(LocationLatitude),
    LocationLongitude = as.numeric(LocationLongitude)
  )


p_load(ggplot2, sf, rnaturalearth, rnaturalearthdata)

# Load world map
world <- ne_countries(scale = "medium", returnclass = "sf")

# Define Finland's bounding box
finland_xlim <- c(19, 32)  # Approximate longitude range of Finland
finland_ylim <- c(59.5, 70)  # Approximate latitude range of Finland

# Create the map
ggplot() +
  geom_sf(data = world, fill = "gray90", color = "gray50") +
  geom_point(data = dat.t, aes(x = LocationLongitude, y = LocationLatitude),
             color = "red", alpha = 0.6, size = 2) +
  coord_sf(xlim = finland_xlim, ylim = finland_ylim) +
  labs(title = "Survey Respondent Locations",
       x = "Longitude", y = "Latitude") +
  theme_minimal()
```

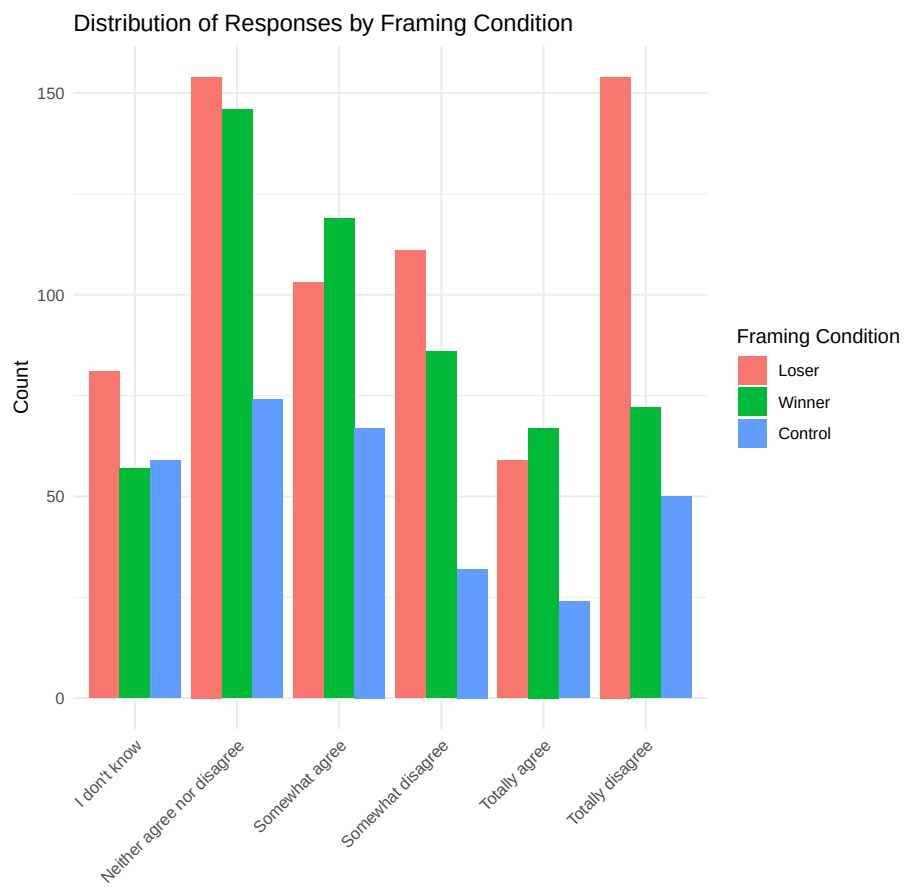



```
dat.t$Priming <- factor(ifelse(is.na(dat.t$DisplayOrder), "Control", dat.t$DisplayOrder))
p_load("tidyverse")
dat.t$Priming <- recode_factor(dat.t$DisplayOrder,
                               `Q25` = "Loser",
                               `Q17` = "Winner",
                               .default = "Control",
                               .missing = "Control")


# plots
p_load(tidyr, ggplot2)

ggplot(dat.t, aes(x = Q13_1, fill = Priming)) +
  geom_bar(position = "dodge") +
  labs(title = "Distribution of Responses by Framing Condition",
```

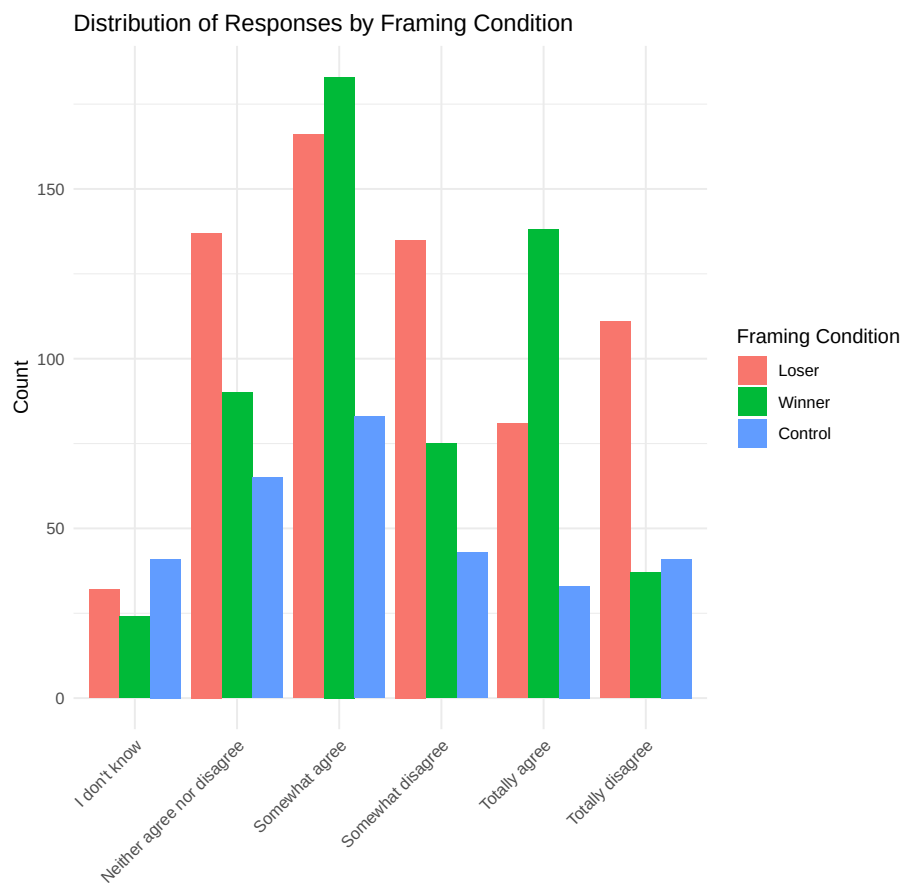
```
       x = "The Gov't must be able to censor media that are unduly critical of the Gov't",
       y = "Count",
       fill = "Framing Condition") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



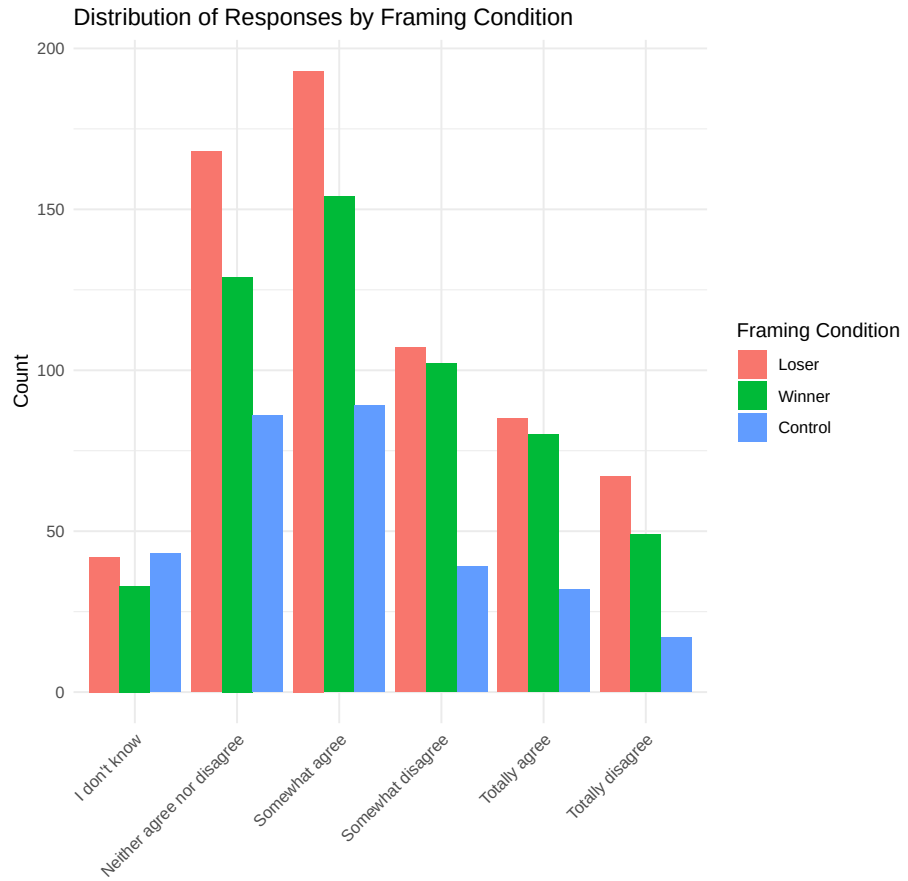

```
ggplot(dat.t, aes(x = Q13_3, fill = Priming)) +
  geom_bar(position = "dodge") +
  labs(title = "Distribution of Responses by Framing Condition",
       x = "The government should be able to ignore court decisions that it considers politi
       y = "Count",
       fill = "Framing Condition") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```




```
ggplot(dat.t, aes(x = Q13_5, fill = Priming)) +
  geom_bar(position = "dodge") +
  labs(title = "Distribution of Responses by Framing Condition",
       x = "Government should be able to stretch the limits of legislation to solve pressing
       y = "Count",
       fill = "Framing Condition") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```




```
ggplot(dat.t, aes(x = Q13_6, fill = Priming)) +
  geom_bar(position = "dodge") +
  labs(title = "Distribution of Responses by Framing Condition",
       x = "Our country would work better if political decisions were left to independent, n
       y = "Count",
       fill = "Framing Condition") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

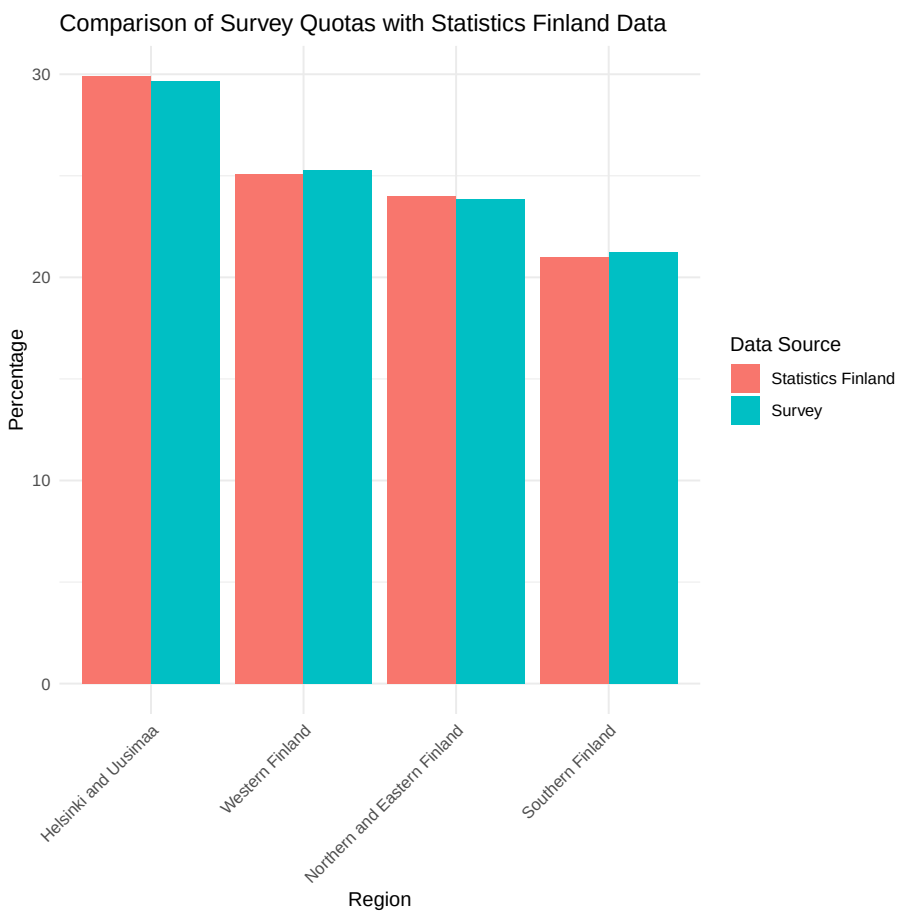



uld work better if political decisions were left to independent, non−elected experts instead of politicians

Sampling

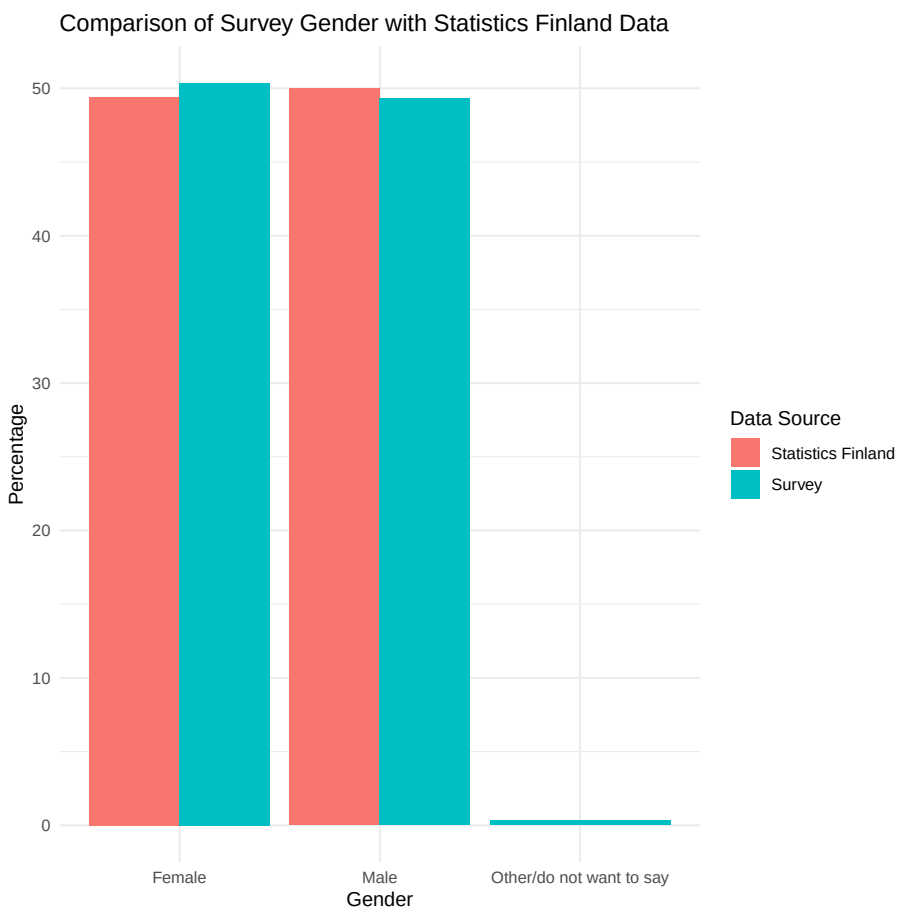
```
# Quota plots

## REGION
fi.eurostat.region <- data.frame(
  Region = c(
    "Helsinki and Uusimaa",
    "Northern and Eastern Finland",
    "Southern Finland",
    "Western Finland"),
  Population_Percentage = c(29.9,24.0,21.0,25.1)
)

# Survey data
fi.survey.data.region <- dat.t %>%
```

```r
  count(Q5) %>%
  mutate(Survey_Percentage = (n / sum(n)) * 100)

# Combine the data
fi.combined.data.region <- fi.eurostat.region %>%
  left_join(fi.survey.data.region, by = c("Region" = "Q5")) %>%
  select(Region, Population_Percentage, Survey_Percentage)

# Correct pivot_longer with updated column names
fi.plot.data.region <- fi.combined.data.region %>%
  pivot_longer(cols = c("Population_Percentage", "Survey_Percentage"),
               names_to = "Source",
               values_to = "Percentage") %>%
  mutate(Source = recode(Source,
                         "Population_Percentage" = "Statistics Finland",
                         "Survey_Percentage" = "Survey"))

# Plot the data
ggplot(fi.plot.data.region, aes(x = reorder(Region, -Percentage), y = Percentage, fill = Sou
  geom_bar(stat = "identity", position = "dodge") +
  labs(title = "Comparison of Survey Quotas with Statistics Finland Data",
       x = "Region",
       y = "Percentage",
       fill = "Data Source") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```




```r
## Gender

p_load(dplyr, ggplot2)

#
fi.survey.gender <- dat.t %>%
  group_by(Q4) %>%
  summarise(Count = n()) %>%
  mutate(Percentage = (Count / sum(Count)) * 100)

#
fi.eurostat.gender <- data.frame(
  Gender = c("Male", "Female"),
  Percentage = c(50.0, 49.4),
  Source = "Statistics Finland"
)
```

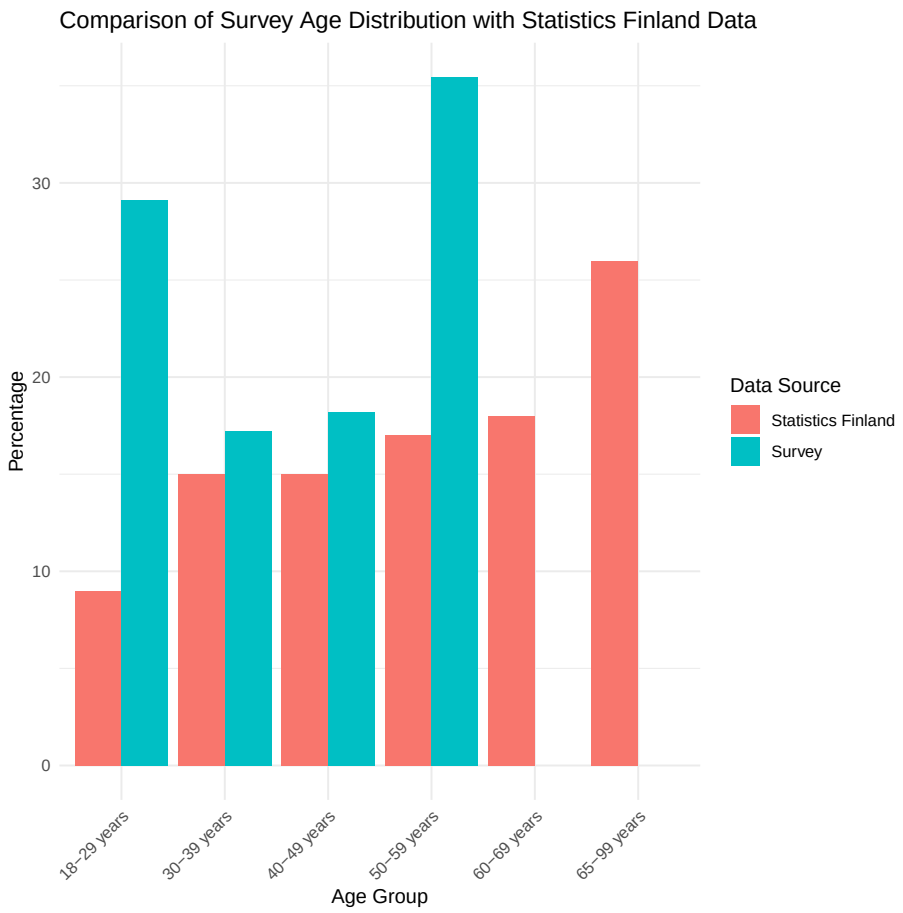
```r
#
fi.survey.gender <- fi.survey.gender %>%
  rename(Gender = Q4) %>%
  mutate(Source = "Survey") %>%
  select(Gender, Percentage, Source)

combined_gender_data.fi <- bind_rows(fi.eurostat.gender, fi.survey.gender)


# Create comparison plot
ggplot(combined_gender_data.fi, aes(x = Gender, y = Percentage, fill = Source)) +
  geom_bar(stat = "identity", position = "dodge") +
  labs(title = "Comparison of Survey Gender with Statistics Finland Data",
       x = "Gender",
       y = "Percentage",
       fill = "Data Source") +
  theme_minimal()
```




```
# Age

## Age
p_load(dplyr, ggplot2, tidyr)

# Statistics Finland age distribution
fi.eurostat.age <- data.frame(
  Age_Group = c("18-29 years", "30-39 years", "40-49 years",
                "50-59 years", "60-69 years", "65-99 years"),
  Population_Percentage = c(9.00, 15.00, 15.00, 17.00, 18.00, 26.00)
)

# Group survey responses to match Statistics Finland's age categories
fi.survey.data.age <- dat.t %>%
  mutate(Age_Group = case_when(
    Q3 %in% c("18-24", "25-34") ~ "18-29 years",  # Merge 18-24 and 25-34 into 18-29
```

```r
    Q3 == "35-44" ~ "30-39 years",  # Adjusting based on available categories
    Q3 == "45-54" ~ "40-49 years",
    Q3 == "55+" ~ "50-59 years",  # Assuming 55+ mostly falls into 50-59
    TRUE ~ NA_character_  # Catch-all for unexpected values
  )) %>%
  filter(!is.na(Age_Group)) %>%
  count(Age_Group) %>%
  mutate(Survey_Percentage = (n / sum(n)) * 100)

# Merge survey and Statistics Finland data
fi.combined.data.age <- fi.eurostat.age %>%
  left_join(fi.survey.data.age, by = "Age_Group") %>%
  select(Age_Group, Population_Percentage, Survey_Percentage)

# Convert to long format for plotting
fi.plot.data.age <- fi.combined.data.age %>%
  pivot_longer(cols = c("Population_Percentage", "Survey_Percentage"),
               names_to = "Source",
               values_to = "Percentage") %>%
  mutate(Source = recode(Source,
                         "Population_Percentage" = "Statistics Finland",
                         "Survey_Percentage" = "Survey"))

# Reorder Age_Group for proper plotting order
fi.plot.data.age <- fi.plot.data.age %>%
  mutate(Age_Group = factor(Age_Group,
                            levels = c("18-29 years",
                                       "30-39 years",
                                       "40-49 years",
                                       "50-59 years",
                                       "60-69 years",
                                       "65-99 years")))

# Generate comparison plot
ggplot(fi.plot.data.age, aes(x = Age_Group, y = Percentage, fill = Source)) +
  geom_bar(stat = "identity", position = "dodge") +
  labs(title = "Comparison of Survey Age Distribution with Statistics Finland Data",
       x = "Age Group",
       y = "Percentage",
       fill = "Data Source") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))

## Warning:  Removed 2 rows containing missing values or values outside
the scale range
```

```
## (`geom_bar()`).
```


Comparison of Survey Age Distribution with Statistics Finland Data
Percentage
0
10
20
30
18–29 years
30–39 years
40–49 years
50–59 years
60–69 years
65–99 years
Age Group
Data Source
Statistics Finland
Survey